The Joy of Computing using Python

Assignment 2

NOTE: Python 3.7 has been used for this Assignment

1. Identify the statement with an invalid syntax. Expected Output: Stay Safe! Friends

   (a) print("Stay Safe!","Friends")

   (b) print("Stay Safe! Friends")

   (c) print('Stay Safe! Friends')

   (d) print(Stay Safe Friends)

   CORRECT ANSWER: (d)

2. Predict the output for the following code that checks the eligibility to vote.

```
age=19
if age>=18
        print('Hey! You are eligible to vote')
else:
        print('OOPS! You are not eligible to vote')
```

   (a) Hey! You are eligible to vote

   (b) OOPS! You are not eligible to vote

   (c) Syntax Error: invalid syntax

   (d) Invalid age

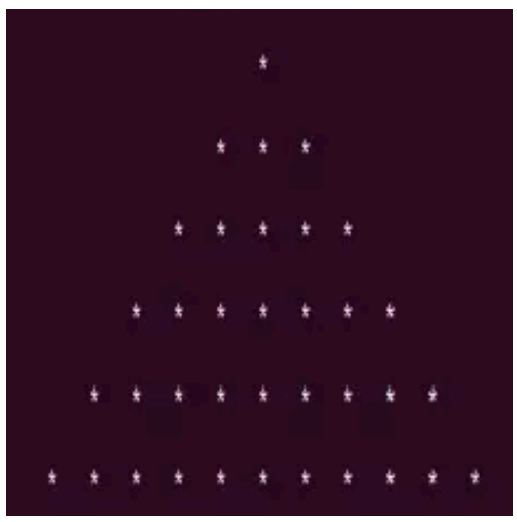
   CORRECT ANSWER: (c)

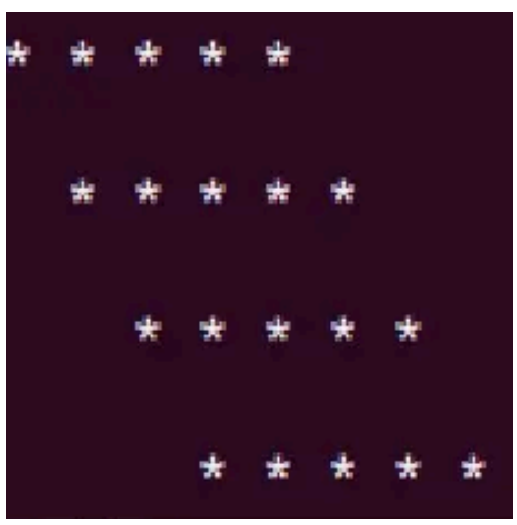
   The option Syntax Error: invalid syntax is correct because colon is missing in the if statement

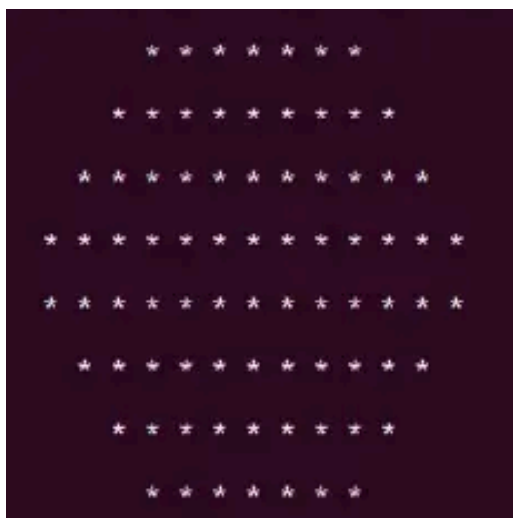
3. What is the output of the following code?

```
def func():
        print()
        c=10
        i=0
        while(i<=5):
                j=1
                while(j<=20):
                        print(' ',end=' ')
                        if(j>=10-i and j<=10+i):
                                print('*',end=" ")
                        else:
                                print(' ',end=" ")
                        j=j+1
                print('\n')
                i=i+1
func()
```

(a) 

(b) 

(c) 

(d) 

CORRECT ANSWER: (b)

4. With n as input, the code below computes

```
def mul(num):
        if(num==1):
                return(-1)
        return(-1*mul(num-1))
```

```
n=int(input("Enter the value of n"))
print(mul(n))
```

(a) $-1 \times n$

(b) $-1 + n$

(c) $(-1)^n$

(d) $n^{(-1)}$

CORRECT ANSWER: (c)

5. For the given code, what is the value of 'total' variable at the end of execution?

```
m1 = '98'
m2 = '79'
m3 = '87'
total = m1 + m2 + m3
print(total)
```

(a) syntax error

(b) 264

(c) 988779

(d) 987987

CORRECT ANSWER: (d)

The last option is correct because the operator '+' concatenates the string inputs.

6. Replace the given set of instructions with a for loop.

```
n=2
print(n)
n=n*2
print(n)
n=n*2
print(n)
n=n*2
print(n)
n=n*2
print(n)
```

(a)
```
for i in range(1,6):
print(pow(2,i))
```

(b)
```
for i in range(1,6):
    print(pow(i,2))
```

(c)
```
for i in range(1,6):
    print(pow(2,i))
```

(d)
```
for i in range(1,5):
    print(pow(2,i))
```

CORRECT ANSWER: (c)

7. Which of the following is the output for the given code?

```
n=5;print(n+5);print(n+5);print(n+5);print(n+5);
```

(a) 10 10 10 10

(b) 5 10 15 20

(c) 10
10
10
10

(d) 5
10
15
20

CORRECT ANSWER: (c)

8. Identify the appropriate output.

```
name='Avani Chaturvedi!'
print('Hello!', name, 'How are you?')
print('Proud to meet you!')
```

(a) Hello! Avani Chaturvedi! How are you? Proud to meet you!

(b) Hello! Avani Chaturvedi!
How are you?
Proud to meet you!

(c) Hello!
Avani Chaturvedi!
How are you?
Proud to meet you!

(d) Hello! Avani Chaturvedi! How are you?
Proud to meet you!

CORRECT ANSWER: (d)

9. What happens if we key in number 5 for the variable c in the below code?

```
c=1
while(c==1):
        print('hello')
        c=int(input('Enter choice:0/1:'))
```

—

(a) Loop terminates

(b) Continues execution

(c) Program restarts execution

(d) None of the above

CORRECT ANSWER: (a)

Loop terminates is the correct answer because the loop continues only when c is equal to 1. For all other values it terminates.

10. What is the output of the following code?

```
d=-20
while(d>0):
    print(d)
if(d<0):
    print(-1*d)
```

(a) Infinite loop

(b) -20 20

(c) 20

(d) -20

CORRECT ANSWER: (c)